

SPEC-DRIVEN COGNITIVE SCAFFOLDING

v1.4.1

Deterministic Architecture for Autonomous Coding Agents

Transitioning AI coding agents from probabilistic prompt-and-pray loops into deterministic cybernetic systems through externalized memory and cryptographic test verification.

Standard: SPEC-001

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The Failure of Monolithic Prompting

Autonomous agents typically fail due to Context Conflation—cramming invariant rules, temporary thoughts, directory structures, and git history into a single bloated context window. This triggers three predictable pathologies:

Pathology >_	The Root Cause	The Result
Context Drift	Overloaded context windows push architectural constraints out of attention.	The agent breaks rules, hallucinates dependencies, and degrades constraints.
Cartographic Hallucination	Lack of an explicit, persistent repository map.	Token-wasting brute-force grep loops and invention of phantom paths.
Episodic Amnesia	No structured negative memory or rejection ledgers.	Multi-turn regression loops re-attempting previously rejected hypotheses.

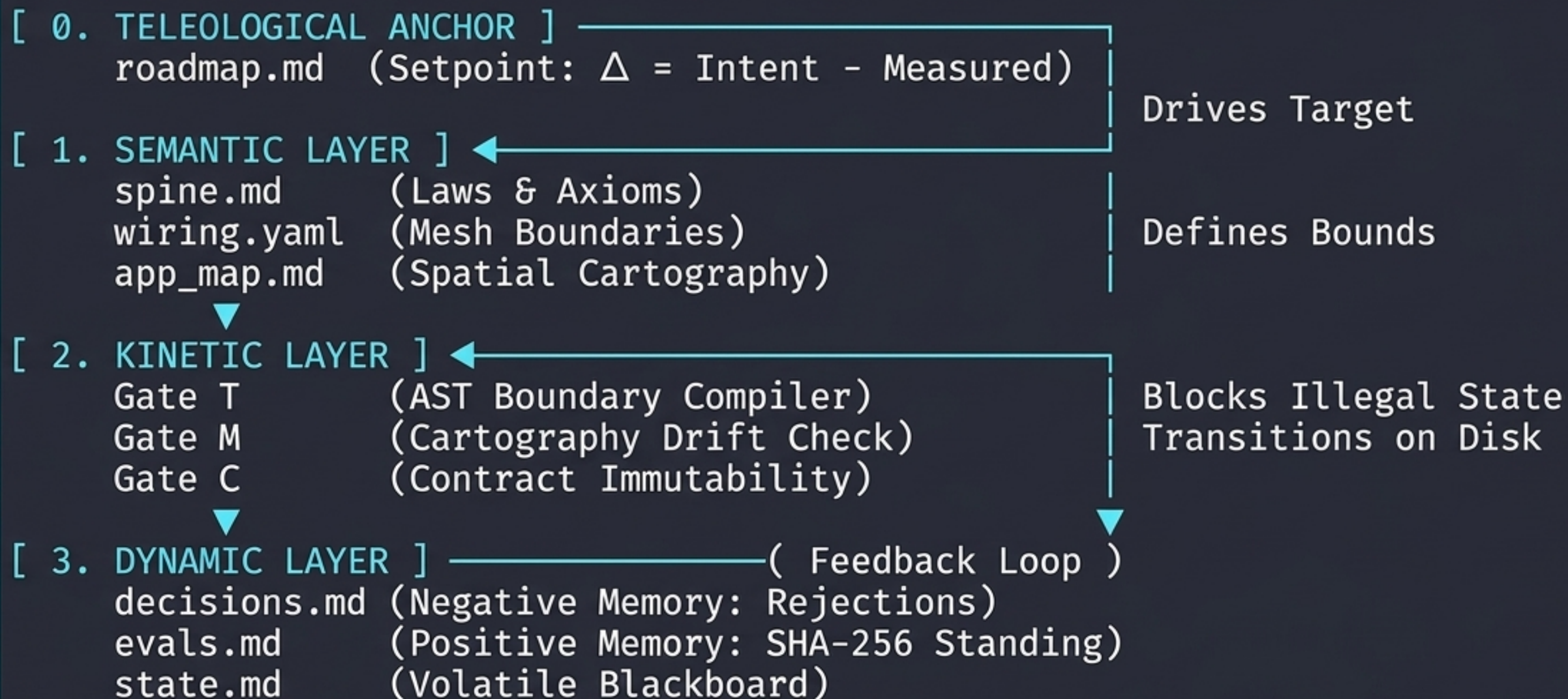
A 7-Pillar Cognitive Topography

SDCS decouples agent cognition into purpose-built, version-controlled artifacts to eliminate context conflation.

#	Artifact	Pillar Role	Analogy	Description
1	spine.md	Constitutional Invariants	The Law	Immutable domain axioms and forbidden actions.
2	wiring.yaml	Declarative Topology	The Mesh	Component boundaries and interface bindings.
3	roadmap.md	Acceptance Contract	The North Star	High-level [INTENT] vs empirical [MEASURED].
4	state.md	Dynamic Working Memory	The Whiteboard	Active objective & blockers. Strict ≤300 token cap.
5	app_map.md	Repository Cartography	The Compass	Structural index consulted prior to file access.
6	decisions.md	Negative Episodic Memory	The Graveyard	Rejection log (Claim → Measurement → Reopen).
7	evals.md	Positive Episodic Memory	The Scorecard	SHA-256 locked benchmark fixtures.

The Operational Ontology

The codebase is modeled as a deterministic cybernetic control system, not a text corpus.



Boundary & Topology Enforcement

Constraining the agent's operational blast radius requires explicit, machine-readable limits.

spine.md (Constitutional Invariants)

- Non-negotiable domain axioms.
- Enforced by Gate C: Commits mutating invariants are physically rejected unless `SDCS_ALLOW_CONSTITUTIONAL_MUTATION=1` is explicitly set by humans.

wiring.yaml (Declarative Topology)

- Maps the service mesh, subsystem interfaces, and package dependency graphs.
- Eliminates circular imports.
- Enforced by Gate T: AST static import auditor runs on pre-commit. Illegal component calls are physically blocked before execution.

The Macro Acceptance Contract

roadmap.md prevents Specification Drift by decoupling optative intent from indicative reality.

The Strict Dichotomy:

1. [INTENT]: Verbatim human requirements. Unalterable by the agent.
2. [MEASURED]: Empirical telemetry. Must link to a verifiable commit hash.

Milestone M-004: Vector Search Subsystem

- * [INTENT]: Sub-50ms p99 latency across 100k indexed vectors under 500 RPS load. Zero disk spillover.
- * [MEASURED]: 41.2ms p99 latency at 500 RPS (Commit a9f4c21); memory ceiling holds at 384MB. Gate PASSED.

Note: The roadmap carries strictly NO ephemeral task queues. Those belong in state.md.

Symmetric Episodic Memory

Agents must know what failed just as clearly as what passed.

Negative Memory (decisions.md)

The **Rejection Graveyard** stops multi-turn regression loops. Forced 3-part schema:

1. **The Claim:** The attempted hypothesis.
2. **The Measurement:** The empirical failure.
3. **Reopen Condition:** Falsifiable trigger required to try again.

Positive Memory (evals.md)

Defeats the **Phantom Corpus** trap where agents fake 100% test passes.

- Cryptographically anchors test fixtures via normalized SHA-256 digests.
- Enforces diversity; halts execution if duplicate byte-identical files are detected.

The Flight Recorder Invariant

To prevent context exhaustion, working memory and historical shift archives are physically separated. Historical session logs are **NEVER** ingested on **Turn 1 boot**.

Dimension	state.md (The Blackboard)	sessions/*.md (Flight Recorder)
Lifecycle	Highly Volatile. Continually overwritten.	Immutable. Write-once, append-only.
Token Budget	Strict ≤ 300 Tokens. Checked via Gate A.	Unbounded historical handoff prose.
Boot Behavior	Always Read. Tells agent current exact state.	Never Read. Stops episodic amnesia.
Structure	Objective, Blockers, Pending Gates.	Shift logs, verification status, handoffs.

Forensic retrieval: Query the single-line JSON index without breaking the invariant:
> `sdcs session list --query "GPU"`

The 4-Phase Autonomous Execution Engine

The agent executes every turn through a deterministic finite state machine, mapping actions directly to the cybernetic ontological layers.

[1. ORIENTATION]

Hydrate state in exact sequence (Pillars 1->3->5->6->7->4).

↳ Semantic & Dynamic Layers establishing Ground Truth.



[2. PLANNING]

Formulate diffs against state.md. Map specific files via app_map.md.

↳ Semantic Layer (wiring.yaml) validates subsystem boundaries.



[3. EXECUTION]

Mutate code. Execute kinetic test gates (Gate T, Gate C, Gate M).

↳ Kinetic Layer physically blocks unauthorized operations.



[4. CLOSE-OUT]

Prune state.md (≤ 300 tokens). Record Rejections. Update roadmap.

↳ Dynamic Layer updates closed-loop feedback array.

Kinetic Defenses in Action

SDCS agents do not debate test failures. They respond to physical laws on disk enforced by automated integrity gating.

```
$ sdcs verify --topology --append-rejections
```

```
[Gate T] Auditing Abstract Syntax Tree imports against wiring.yaml...
```

```
✗ VIOLATION: src/api/auth.py:42
```

```
  Claim: 'api' subsystem can import 'db_core' directly from 'storage'.
```

```
  Measurement: AST Static Analysis rejected prohibited module boundary.
```

```
↳ Serializing Inverted ADR (## REJ-014) to decisions.md...
```

```
↳ Aborting transaction. Agent must use dependency inversion to proceed.
```

- Gate M (Drift): Blocks commits if agent creates files unindexed in app_map.md.
- Gate C (Immutability): Rejects unauthorized mutation of constitutional invariants.
- Gate A (Budget): sdcs verify --state fails if whiteboard exceeds 350 tokens.

Lossless Compaction Protocol

When an agent hits token saturation (~80%), context reset induces Compaction Amnesia. SDCS solves this with a standardized 5-Step Checkpoint Protocol.



Post-compact Turn 1 boot reads the pristine 300-token `state.md`. Compaction amnesia is eliminated.

Deploying Deterministic Agents

Standardize your agentic loops into deterministic systems today. SDCS profiles adapt to **Lite** (~400 token) or **Full Shift** (~3,500 token) engineering workloads.

1. Scaffold the 7-Pillar Architecture

```
pip install sdcs
sdcs init --use-agent-dir
```

2. Audit Drift & Recalibrate Golden Fixtures

```
sdcs map --check
# Verify cartography matches
physical disk

sdcs eval record all
# Atomically compute SHA-256
for all test fixtures
```

3. Enforce Codebase Invariants (CI/CD or Git Hooks)

```
sdcs verify --all
# Runs Gate T (Topology),
Gate A (State), and Evals
```